

A Digital Superresolution Method With Minimal Sensitivity to Shift Estimation Error

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Abstract—Superresolution (SR) methods become essential when an undersampled low-resolution (LR) image is unable to provide accurate target detection. The estimation of an HR image from a single LR image is ill-posed problem, and hence requires prior information. More the constraints, better is the reconstruction accuracy and this forms the basis of most of the contemporary state-of-the-art (SOTA) superresolution methods such as SRCNN and SRGAN which implement prior information as training set. Yet another approach to overcome the ill-posedness of the problem is to have multiple diverse LR images with the potential to reconstruct accurate HR image. Here we present the performance analysis of a generalized sampling theorem (GST) based multi-frame SR method. A simulation study using Gaussian targets is conducted, and a comparative performance analysis of the GST multi-frame SR method with the traditional multi-frame interpolation schemes and SOTA methods is presented using the percentage mean square error (%MSE) and Structural Similarity Index Measure (SSIM). Our findings indicate that the GST SR method can outperform traditional interpolation and the SOTA methods.

Index Terms—Gaussian target, interpolation, MSE, SSIM, superresolution (SR).

I. INTRODUCTION

DIGITAL superresolution (DSR) is commonly used in modern imaging systems to reconstruct high resolution (HR) image from undersampled low-resolution (USLR) image frames, where the undersampling is caused due to the large pixel-size in the sensor array recording the image [1], [2]. The problem of estimating an HR image from a single LR image is an ill-posed problem, and hence requires prior information (implemented as constraints) to help with increasing the estimation accuracy. This approach forms the backbone of the current state-of-the-art (SOTA) SR methods such as Super-resolution Convolutional Neural Network (SRCNN) and Super-resolution Generative Adversarial Networks (SRGAN) [3], [4], [5], [6], [7]. Though these methods have shown impressive results, but high computational cost and dependency on large amounts of high-quality training data, limit their applicability in many of the real-world practical scenarios. A different approach to estimate an HR image involves stitching together, using varied

spatial interpolation schemes [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], of multiple diverse LR image frames acquired with sub-pixel translational shifts of the sensor array with-in the focal plane. Such techniques are extensively used in target detection applications for satellite imagery [19], [20], medical diagnosis [21], [22], [23], surveillance [24], microgrid polarimeter imaging [25], etc to increase image resolution for target recognition purposes. But this approach does not provide accurate reconstructions for all class of objects, and its reconstruction accuracy suffers dramatically with any error in shift estimation. Clearly a robust mathematically rigorous method with the potential to provide accurate estimate in the presence of errors in shift estimates is needed here. The generalized sampling theorem [26], [27] provides a way to mathematically formulate the estimation of an HR image from multiple diverse USLR images as a well-posed problem with a unique solution. The diversity between multiple data sets could be introduced using sub-pixel shifts, tilt or defocus and LR frames could be stitched together using the theoretical framework of the GST. We call this method as GST DSR approach.

In this work, we conduct a study of the GST DSR method based on multiple sub-pixel shifted USLR frames acquired via translational shifts of the sensor array with-in the focal plane. A complete theoretical framework, and a comparative performance analysis with the conventional interpolation based methods and some of the SOTA methods are presented. Since errors in estimation of shift between the multiple LR frames is inevitable in most practical scenarios, primarily those involving moving platforms such as a satellite, our study emphasizes on the robustness of the methods to the frame-to-frame shift estimation errors.

We provide the mathematical framework of the 1D GST DSR technique in Section II and then extend it to 2D. We evaluate the GST DSR technique using synthetic data and compare it against a conventional interpolation approach and SOTA methods in Section III. Conclusions, limitations, and future applications of this work are discussed in Section IV.

II. PROPOSED GST DSR METHOD

Consider a 1D HR scene $g(x)$ which is continuous, $(-B_x, B_x)$ band-limited, and it is sampled with point-sized pixels with an undersampling factor of $L_x = 2$, which corresponds to a sampling interval of $\frac{2}{2B_x}$. Two USLR frames are obtained by shifting the pixel array by half of the sampling interval, as illustrated in Fig. 1. For such a sequence, notice that the critically sampled HR scene can be obtained by simply inserting the samples of the second image in between the samples of the first image. In practice, controlling the frame-to-frame shift with such precision is difficult, and hence there is a need to allow for an arbitrary sub-pixel shift. Here we present a GST based

Received 30 May 2024; revised 7 August 2024; accepted 4 September 2024. Date of publication 23 September 2024; date of current version 8 October 2024. The associate editor coordinating the review of this article and approving it for publication was Dr. Sambit Bakshi. (Corresponding author: Neethu S. Ravi.)

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Digital Object Identifier 10.1109/LSP.2024.3466008

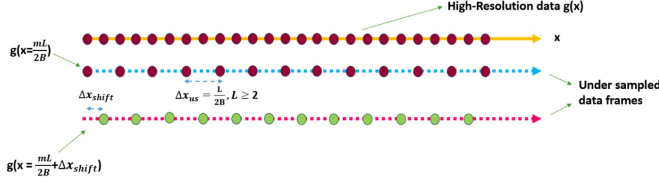


Fig. 1. Illustration of the undersampling process in 1D using point-sampling.

mathematical analysis to stitch multiple LR frames together in order to estimate an HR image. The bandlimited HR scene $g(x)$ can be written as

$$g(x) = \int_{-B_x}^{B_x} G(v_x) e^{j2\pi v_x x} dv_x, \quad (1)$$

where $G(v_x)$ is the Fourier Transform of $g(x)$. By dividing the frequency range from $-B_x$ till B_x with a step size of $\frac{2B_x}{L_x}$, we obtain

$$g(x) = \int_{-B_x}^{-B_x + \frac{2B_x}{L_x}} G(v_x) e^{j2\pi v_x x} dv_x \dots \dots + \int_{B_x - (L_x - 1) \times \frac{2B_x}{L_x}}^{B_x} G(v_x) e^{j2\pi v_x x} dv_x. \quad (2)$$

For the special case of undersampling factor $L_x = 2$, the under-sampled data will be given as $g(x = \frac{mL_x}{2B_x}) = g(\frac{m}{B_x})$, $m \in I$, and is expressed as

$$\begin{aligned} g\left(\frac{m}{B_x}\right) &= \int_{-B_x}^0 G(v_x) e^{j2\pi(\frac{m}{B_x} v_x)} dv_x \\ &\quad + \int_0^{B_x} G(v_x) e^{j2\pi(\frac{m}{B_x} v_x)} dv_x \\ &= \int_{-B_x}^0 \left[G(v_x) + G(v_x + B_x) \right] e^{j2\pi(\frac{m}{B_x} v_x)} dv_x \\ &= \int_{-B_x}^0 J(v_x) e^{j2\pi(\frac{m}{B_x} v_x)} dv_x \end{aligned} \quad (3)$$

where $J(v_x) \equiv G(v_x) + G(v_x + B_x)$. Now, consider a periodic function $J'(v_x)$, with periodicity B_x , such that $J'(v_x) = J(v_x)$, $-B_x < v_x < 0$, and hence can be expressed as a Fourier series

$$J'(v_x) = \sum_{m=-\infty}^{\infty} \phi(m) e^{-j2\pi(v_x \frac{m}{B_x})}, m \in I \quad (4)$$

where the Fourier series coefficients $\phi(m)$ are given by

$$\begin{aligned} \phi(m) &= \frac{1}{B_x} \int_{-B_x}^0 J'(v_x) e^{j2\pi(v_x \frac{m}{B_x})} dv_x \\ &= \frac{1}{B_x} g\left(\frac{m}{B_x}\right) \end{aligned} \quad (5)$$

Here (5) is obtained using (3). Using Eqns 4 and 5, we obtain

$$\begin{aligned} G(v_x) + G(v_x + B_x) &= \frac{1}{B_x} \sum_{m=-\infty}^{\infty} g\left(\frac{m}{B_x}\right) e^{-j2\pi(v_x \frac{m}{B_x})} \\ \forall -B_x \leq v_x \leq 0 \end{aligned} \quad (6)$$

Since we have two unknowns $G(v_x)$ and $G(v_x + B_x)$, but only one equation, we need to have another linearly independent equation to be able to evaluate $G(v_x)$ uniquely, and this is provided by a shifted USLR frame, with a Δx sub-pixel shift between the two frames.

$$g\left(\frac{m}{B_x} + \Delta x\right) = \int_{-B_x}^{B_x} G(v_x) e^{j2\pi v_x (\frac{m}{B_x} + \Delta x)} dv_x. \quad (7)$$

Carrying a similar analysis as we did for the first frame, the problem could be eventually formulated in a matrix equation as

$$\begin{aligned} \begin{pmatrix} 1 & 1 \\ e^{j2\pi v_x \Delta x} & e^{j2\pi(v_x + B_x)\Delta x} \end{pmatrix} \begin{pmatrix} G(v_x) \\ G(v_x + B_x) \end{pmatrix} \\ = \begin{pmatrix} \sum_{m=-\infty}^{\infty} \frac{1}{B_x} g\left(\frac{m}{B_x}\right) e^{-j2\pi(v_x \frac{m}{B_x})} \\ \sum_{m=-\infty}^{\infty} \frac{1}{B_x} g\left(\frac{m}{B_x} + \Delta x\right) e^{-j2\pi(v_x \frac{m}{B_x})} \end{pmatrix} \end{aligned} \quad (8)$$

Thus, by a proper choice of Δx so that the system matrix is invertible, we can uniquely estimate the Fourier transform $G(v_x)$, and finally the HR spatial image is obtained via the inverse Fourier transform. We can follow a similar development to extend the DSR approach to 2D where four under-sampled, shifted frames are used for HR reconstruction. Here the 2D scene $g(x, y)$ is assumed continuous and bandlimited in the range of $\{v_x, v_y\} \in \{(-B_x, B_x), (-B_y, B_y)\}$. The resulting set of linearly independent equations for the DSR approach is

III. RESULTS AND DISCUSSIONS

We simulated 1D and 2D data to study the reconstruction ability of the GST DSR algorithm and compared it against traditional interpolation-based approach. The continuous 1D and 2D scenes are assumed to be Gaussian according to

$$g(x) = e^{-\frac{x^2}{\sigma^2}} \quad (10)$$

and

$$g(x, y) = e^{-\frac{x^2 + y^2}{\sigma^2}}, \quad (11)$$

where (x, y) indicate spatial coordinates and σ denotes variance. In each case, a variance of $\sigma = 0.005$ and 0.1 is used for the 1D and 2D scenes, respectively, to yield $B_x = B_y \approx 10$ cycles/length. When $L_x = L_y = 2$, we require two undersampled frames ($g(x = \frac{m}{B_x})$ and $g(x = \frac{m}{B_x} + \Delta x)$) for 1D reconstruction and 4 undersampled frames for 2D reconstruction ($g(x = \frac{m}{B_x}, y = \frac{m}{B_y})$, $g(x = \frac{m}{B_x} + \Delta x, y = \frac{m}{B_y})$, $g(x = \frac{m}{B_x}, y = \frac{m}{B_y} + \Delta y)$, and $g(x = \frac{m}{B_x} + \Delta x, y = \frac{m}{B_y} + \Delta y)$). The critical sampling interval (CSI) $\Delta x_c = \frac{1}{20}$ units and the sampling interval for the undersampled frames is $\Delta x_u = \frac{1}{10}$ units. The required undersampled frames in each case are obtained using shifts of $\Delta x = \Delta y = \frac{1}{20}$ units. 1D and 2D HR scene of size 1×101 and 101×101 are created respectively, and undersampled data of size 1×11 is extracted from it for 1D and size 11×11 for 2D.

A. GST DSR Evaluation

The respective GST DSR algorithms from (8) and (9) showna at the bottom of the next page are applied to these USLR frames to estimate $G(v_x)$ and $G(v_x, v_y)$. The inverse Fourier transform is then applied to reconstruct the HR scene. This technique is illustrated for 1D using 2-frame interpolation (2FI) [8] and for

TABLE I
PERCENTAGE MSE AND SSIM AGAINST GROUND TRUTH FOR GST DSR, 2FI, AND MFI FOR THE 1D AND 2D APPROACHES

Fractional shift Methods	%MSE					SSIM				
	20%	40%	60%	80%	100%	20%	40%	60%	80%	100%
1D GST DSR	4.2e-03	3e-03	2e-03	1e-03	1e-03	0.99	0.99	0.99	0.99	0.99
2FI	0.330	0.0299	0.0232	0.0096	1e-03	0.90	0.93	0.95	0.98	0.99
2D GST DSR	6.64e-09	4.20e-09	2.07e-09	1.04e-09	8.12e-10	1	1	1	1	1
MFI	0.224	0.155	0.087	0.028	6.91e-10	0.74	0.79	0.85	0.93	1

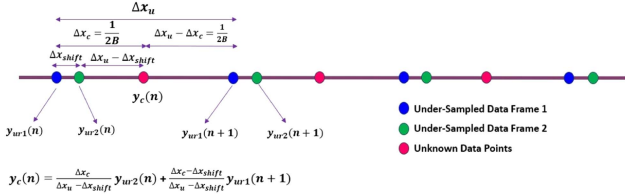


Fig. 2. Illustration of the 1D interpolation approach.

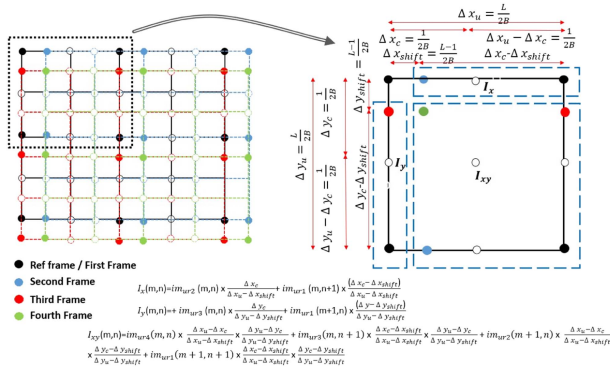


Fig. 3. Illustration of the multi-frame interpolation approach.

2D using multi-frame interpolation (MFI) [8] in Figs. 2 and 3, respectively. It is worth noting that the normalized interpolation weights depend upon the fractional shift applied to the CSI.

We evaluate the resulting reconstructions against the true HR data (ground truth) using %MSE [28] and SSIM metrics [29]. Table I displays results for the 1D and 2D cases for the GST DSR, 2FI, and MFI reconstructions when different subpixel

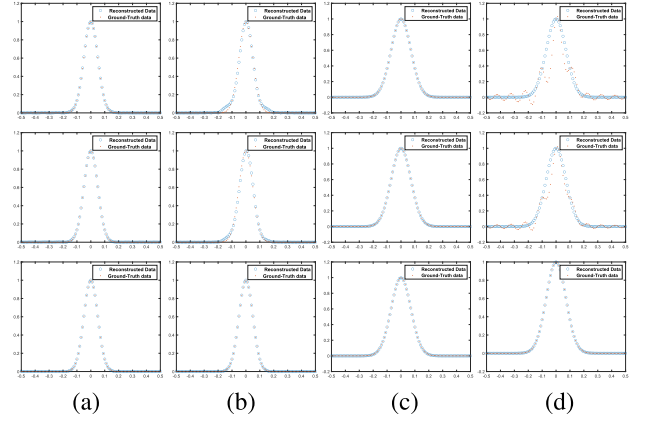


Fig. 4. Reconstruction comparisons of each method against ground truth for the 1D and 2D (taking the 1D cross section) cases for shifts from 20–100% of the CSI. (a) 1D GST DSR, (b) 2FI, (c) 2D GST DSR and (d) MFI approaches.

shifts are used to create the undersampled frames. We define these shifts in terms of percentage of the CSI from 20%–100% and the HR scene is then reconstructed using the GST DSR, 2FI, and MFI algorithms assuming no error in the frame-to-frame shift. These results show that the GST DSR reconstruction is unaffected by the amount of shift, whereas the interpolation-based reconstruction strongly depends on the amount of shift, delivering best estimate at 100% CSI shift. Fig. 4(a), (b), (c) and (d) show comparisons of the reconstructed data against ground truth for the 1D GST DSR, 2FI, 2D GST DSR and MFI methods for fractional shifts of 20%, 60%, and 100% of the CSI, respectively. 1D cross-section is shown for 2D data. Fig. 5 shows the graphical representation of Table I data for fractional

$$\begin{aligned}
 & \begin{pmatrix} 1 & 1 & 1 & 1 \\ e^{j2\pi v_x \Delta x} & e^{j2\pi(v_x+B_x)\Delta x} & e^{j2\pi v_x \Delta x} & e^{j2\pi(v_x+B_x)\Delta x} \\ e^{j2\pi v_y \Delta y} & e^{j2\pi v_y \Delta y} & e^{j2\pi(v_y+B_y)\Delta y} & e^{j2\pi(v_y+B_y)\Delta y} \\ e^{j2\pi(v_x \Delta x + v_y \Delta y)} & e^{j2\pi((v_x+B_x)\Delta x + v_y \Delta y)} & e^{j2\pi(v_x \Delta x + (v_y+B_y)\Delta y)} & e^{j2\pi((v_x+B_x)\Delta x + (v_y+B_y)\Delta y)} \end{pmatrix} \\
 & \times \begin{pmatrix} G(v_x, v_y) \\ G(v_x + B_x, v_y) \\ G(v_x, v_y + B_y) \\ G(v_x + B_x, v_y + B_y) \end{pmatrix} \\
 & = \begin{pmatrix} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g(x = \frac{m}{B_x}, y = \frac{n}{B_y}) e^{-j2\pi(v_x \frac{m}{B_x} + v_y \frac{n}{B_y})} \\ \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g(x = \frac{m}{B_x} + \Delta x, y = \frac{n}{B_y}) e^{-j2\pi(v_x \frac{m}{B_x} + v_y \frac{n}{B_y})} \\ \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g(x = \frac{m}{B_x}, y = \frac{n}{B_y} + \Delta y) e^{-j2\pi(v_x \frac{m}{B_x} + v_y \frac{n}{B_y})} \\ \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g(x = \frac{m}{B_x} + \Delta x, y = \frac{n}{B_y} + \Delta y) e^{-j2\pi(v_x \frac{m}{B_x} + v_y \frac{n}{B_y})} \end{pmatrix}. \quad (9)
 \end{aligned}$$

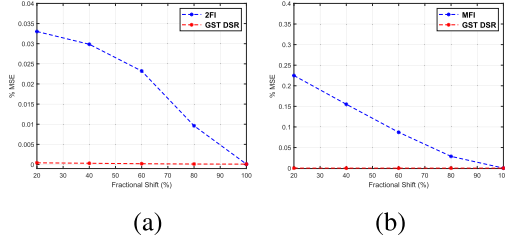


Fig. 5. Reconstruction error for (a) 1D GST DSR and 2FI and (b) 2D GST DSR and MFI as a function of frame-to-frame shift in terms of percentage of the CSI. Reconstructions are performed using zero error in the shift.

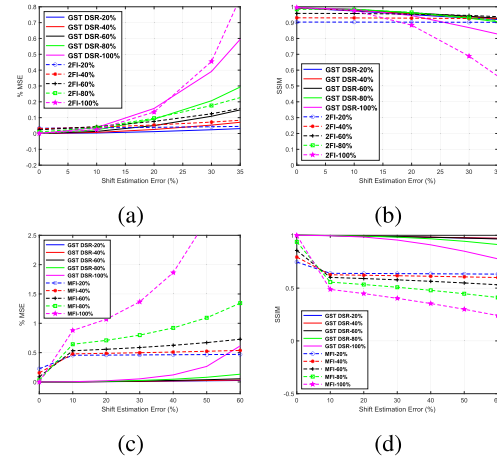


Fig. 6. Reconstruction error comparisons as a function of shift estimation error: (a) 1D GST DSR and 2FI %MSE plot, (b) 1D GST DSR and 2FI SSIM plot, (c) 2D GST DSR and MFI %MSE plot, and (d) 2D GST DSR and MFI SSIM plot.

shifts from (20%–100%) CSI for no error in the shift. These results show that the GST DSR method provides more accurate estimation than interpolation schemes.

B. Shift Estimation Error

In many applications involving moving platforms such as satellite, it is often difficult to accurately control sensor shifts, which lead to error in shift estimation. Consequently, these shift estimation errors impact the quality of HR scene reconstruction. We investigated the effect of shift estimation error on HR scene reconstruction produced by the GST DSR (1D and 2D) and 2FI and MFI approaches for different values of frame-to-frame shifts. Fig. 6 plots reconstruction error for each algorithm against ground truth in terms of %MSE and SSIM. Though reconstruction error rises as shift estimation error increases for all methods, but the rate with which the performance worsens clearly shows that the GST DSR method provides comparatively more robust performance. Fig. 7 provides a visual comparison of 2D reconstructed data using GST DSR and MFI for 0% and 30% shift error for data sampled with a fractional shift of 100% (CSI). Fig. 7(e) particularly shows the detrimental effect of shift estimation error on the MFI approach whereas the corresponding 2D GST DSR reconstruction is visually unaffected.

In a continued effort to study the performance of GST DSR method, the authors conducted a quantitative study of the comparative performance analysis of GST DSR method with single-frame SOTA approaches like SRCNN [5], SRGAN [6],

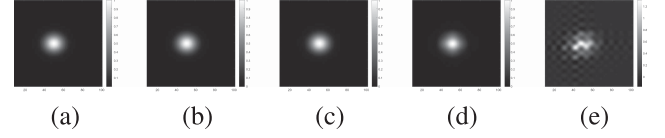


Fig. 7. Visual comparison of HR reconstructed data (a) ground truth, (b) 2D GST DSR for 0% shift error, (c) MFI for 0% shift error, (d) 2D GST DSR for 30% shift error, and (e) MFI for 30% shift error.

TABLE II
MODEL EFFICIENCY ANALYSIS

	% MSE	SSIM	Running Time(s)
MFI [8]	6.919e-10	1	5.57
SRCNN [5]	0.1296	0.852	6.36
SRGAN [6]	2.090	0.473	7.41
ESRGAN [7]	1.772	0.758	12.39
2D GST DSR	8.122e-10	1	7.33

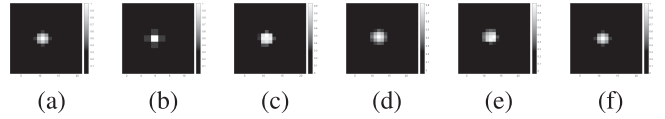


Fig. 8. Visual comparison of GST DSR with SOTA methods. (a) ground truth, (b) USLR data, (c) SRCNN, (d) SRGAN, (e) ESRGAN, and (f) 2D GST DSR for 0% shift error.

and ESRGAN [7] to illustrate its efficacy. By utilizing the source code from the respective references, Table II displays the model efficiency analysis based on %MSE, SSIM, and running time (sec) and the visual comparison of GST DSR with the SOTA methods is shown in Fig. 8. These results indicate that the 2D GST DSR method outperforms the SOTA methods, while the MFI method has a faster running time.

IV. CONCLUSION

In this letter, a rigorous theoretical analysis of GST-DSR method, based on multiple sub-pixel shifted LR frames obtained using translational shifts within the focal plane, is presented. A performance analysis of the method is presented as a comparative study with the conventional interpolation-based methods and the contemporary SOTA methods. Simulation study shows that the reconstruction accuracy in the GST DSR method is more tolerant to frame-to-frame shift estimation error than the conventional interpolation approaches. A characteristic that makes GST DSR method more suitable for real-world practical applications which involve moving platforms such as satellite imaging. In principle, this method could be used for any level of resolution enhancement, but the need to generate larger number of diverse LR frames poses practical challenges. Another limitation of this method is the requirement for prior knowledge of the band-limit of the target, which in many real-world applications, may pose a challenge. The authors would like to point out that even though the interpolation approach lacks in estimation accuracy, but its computation time is lesser, which will make it suitable for applications where estimation accuracy could be compromised for quick estimation. Future work will focus on applying the GST DSR method to microgrid imaging polarimeters to reduce aliasing and IFOV error in sensor data [30], [31] and expand the framework to incorporate camera tilt and defocus diversity.

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